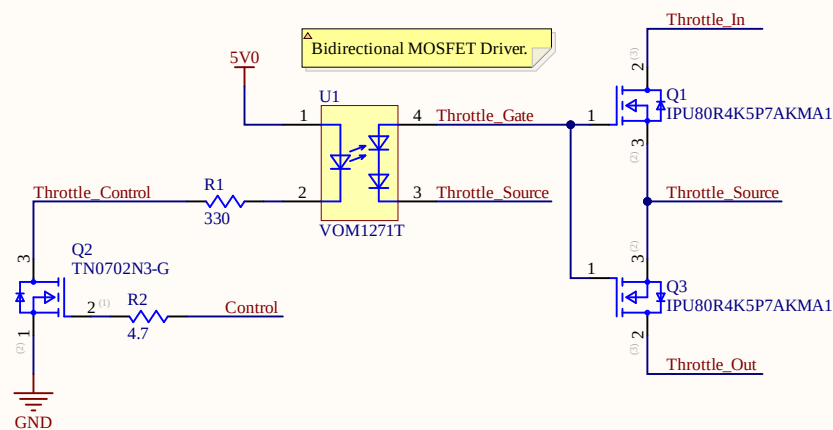
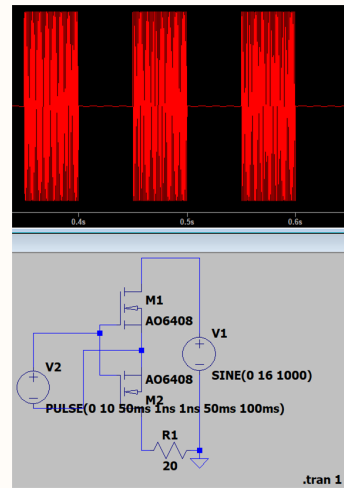
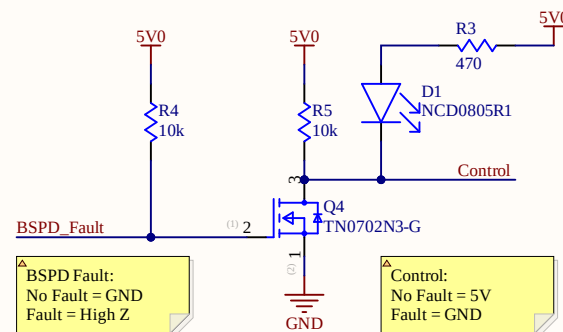




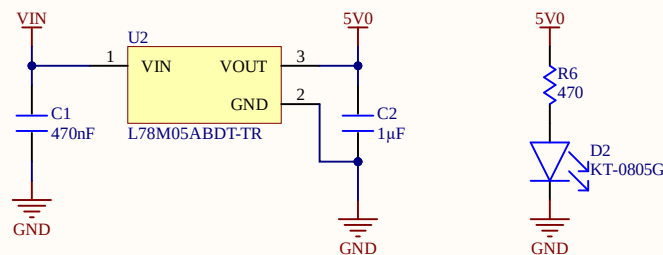
△ MOSFET driver drives 8.7V higher than GND reference. In this case the GND reference is the Source of the Mosfets, This causes the gates to always be higher regardless of AC state. 2 Mosfets prevent body diode stuff.



Bidirectional Switch

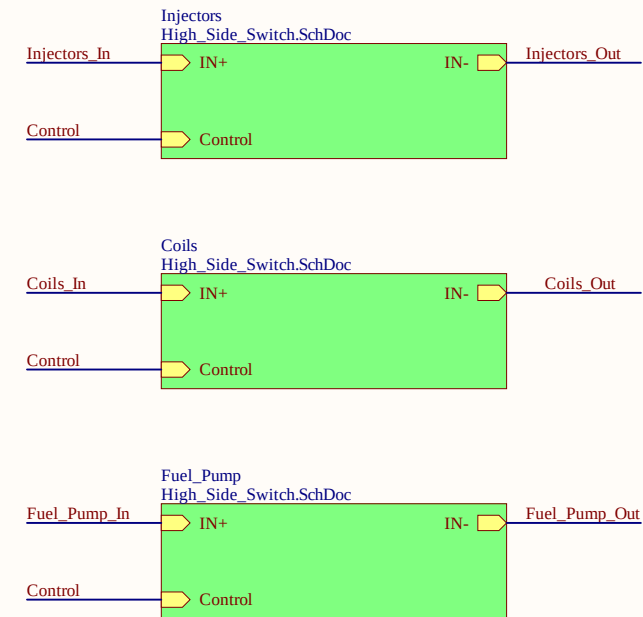


BSPD Fault Translation



Power Budget:
2x LED: 11mA each
VOM1271T: 15mA
Total: 37mA

Power

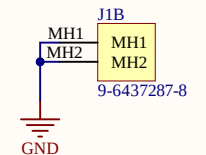


High-Side Switches

Current Sizing:
Fuel Pump: 7A Continuous
Throttle: 4A Continuous
Injectors: 2A Continuous
Coils: 2A Continuous

J1A			
<u>Injectors_In</u>	1	14	<u>Coils_In</u>
	2	15	
	3	16	<u>Injectors_Out</u>
<u>Throttle_Out</u>	4	17	<u>Injectors_Out</u>
<u>Throttle_In</u>	5	18	<u>Injectors_Out</u>
<u>VIN</u>	6	19	<u>Fuel_Pump_Out</u>
<u>BSPD_Fault</u>	7	20	<u>Coils_Out</u>
<u>Injectors_Out</u>	8	21	<u>Coils_Out</u>
<u>Injectors_Out</u>	9	22	<u>Coils_Out</u>
<u>Injectors_Out</u>	10	23	<u>Coils_Out</u>
<u>Injectors_Out</u>	11	24	
<u>Injectors_Out</u>	12	25	
<u>GND</u>	13	26	<u>Fuel_Pump_In</u>

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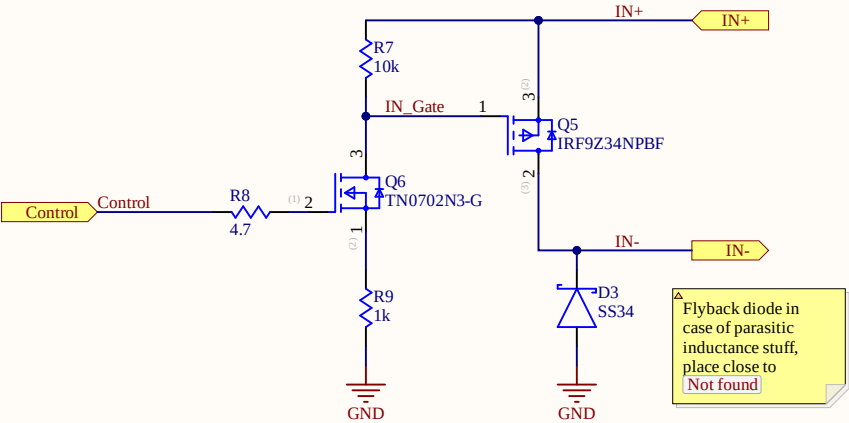
AMPSEAL

PROJECT	BSPD Relay			TITLE	
Last Edit:	8/25/2026			Main	
Date:	8/27/2026			CAR	
Part Num:	ELC-BSPD R-1			MF14 SIZE B REV 1	
Drawn By:	Grant Uher			Mines Formula SAE Electrical	
Approved By:				SHEET NUMBER 1 OF 2	

Control = 0V: NMOS = OPEN, PMOS Gate = IN+, PMOS = OPEN
For IN+ @ 5V
Control = 5V: NMOS = CLOSED, PMOS Gate = 0.45V, PMOS = CLOSED

When NMOS is open, no current goes to GND, so gate is stuck at IN+ voltage.
When NMOS is closed, a voltage divider between 10k and 1k is created which in every case above 2.7V will close the PMOS. $V_{gs(th)}$ is -2.4V.

1k is to prevent NMOS capacitance from discharging too quickly.



Max Current:

Injectors & Coils: 2A Max (according to graphna data prob reality bit higher)

Fuel Pump: Around 5A continuous, 6A max (again according to graphna data prob reality bit higher)

PROJECT		TITLE		
BSPD Relay		High Side Switch		
Last Edit: 8/22/2026		CAR	SIZE	REV
Date: 8/27/2026		MF14	A	1
Part Num: ELC-BSPD_R-1		Mines Formula SAE Electrical		
Drawn By: Grant Uher		SHEET NUMBER 2 OF 2		
Approved By:				

